

Spring 2026
MIS 947/985 Practical Business Analytics

Group Exercise 1

- You can use either R or Python to perform your work.
 - Graded out of 100 points. Please typeset your answers, save answers in R source code or Python .ipynb files with title "_Exercise1_.R/.ipynb" (e.g., Group1_Exercise1.R/Group1_Exercise1.ipynb).
 - Please submit your code to NSYSU Cyber University before **4/23 18:00pm.**
 - Your code must follow the suggested programming and data analysis styles discussed in the class. Please add comments to help understand the code.
- 1. [30 pts] Please load the given heart disease dataset (heart_disease.csv) and answer the following questions.**
- 1.1. [5 pts] Calculate the number of NA values in each column with any "apply" functions, and remove those records (rows) with NA values.
 - 1.2. [6 pts] Create bar charts for categorical variables and box plots for continuous variables. Additionally, calculate the median and quartiles for each continuous variable.
 - 1.3. [5 pts] Please use One-Hot Encoding to convert categorical variables into numerical variables.
 - 1.4. [4 pts] Split the dataset into training (70%) and testing (30%) sets, using random_seed = 2024 to ensure reproducibility and consistent results.
 - 1.5. [6 pts] Build a Logistic Regression model to predict the target variable (Heart Disease Status). Evaluate the model using the AUC on both the training and testing sets.
 - 1.6. [4 pts] Compare the training AUC and testing AUC. Based on the comparison, briefly discuss whether the model shows signs of overfitting or underfitting, and interpret the model's generalizability.

2. **[35 pts] Please load the given subscription revenue dataset (subscription_revenue.csv) and answer the following questions. Assume the target variable is monthly_revenue.**
- 2.1. [7 pts] Please calculate the average monthly_revenue for each combination of contract_type and payment_method, and order the results in descending order of average revenue. Which combination has the highest average revenue?
 - 2.2. [8 pts] Please conduct a series of bivariate analyses on monthly_revenue versus all other variables. Use appropriate plots and statistical tests to examine the relationships. Briefly summarize which variables appear to have the strongest relationships with monthly_revenue.
 - 2.3. [6 pts] Please examine the distribution of monthly_revenue and identify whether severe skewness or extreme values are present. Based on your findings, discuss whether transformation and/or outlier treatment would be reasonable before fitting a linear model.
 - 2.4. [6 pts] Based on your bivariate analyses and distribution findings in Questions 2-2 and 2-3, manually select at least 3 predictor variables and fit a multiple linear regression model to explain monthly_revenue.. Report the variables with p-value < 0.05 and briefly interpret the direction of at least two important coefficients in business terms.
 - 2.5. [8 pts] Please examine correlation and/or multicollinearity among the predictor variables. Report any pairs of variables with high correlations, or variables with high VIF values if applicable. Do you think correlation is a good measurement for variable importance ranking? Briefly explain your answer.
3. **[35 pts] Please load the given customer analytics datasets (customers.csv, orders.csv, and engagement.csv) and answer the following questions. Use “customer_id” as the primary key whenever appropriate.**
- 3.1. [8 pts] Please merge the three datasets into a customer-level analytical table. After merging, check whether there are “duplicated keys”,

“unmatched records”, or “suspicious records”. Briefly report what you found.

- 3.2. [7 pts] Please compute the following business metrics by “acquisition_channel” and “membership_tier”: “average order value”, “purchase frequency”, and “return rate in merchandise”. Which segment appears to be the most valuable, and which segment appears to be the most risky?
- 3.3. [8 pts] Suppose repeat_purchase is the outcome variable. Please conduct a series of bivariate analyses on repeat_purchase versus all other variables. Use appropriate plots, summary tables, and statistical tests to identify variables that may be associated with repeat purchase behavior.
- 3.4. [6 pts] Please create “**at least three**” derived features from the historical customer behavior data, such as recency, frequency, average order value, engagement intensity, or discount usage. Briefly explain why each derived feature may be useful for future predictive analysis.
- 3.5. [6 pts] Based on your previous analyses, please recommend which variables or derived features should be prioritized in a future predictive model for customer retention or upselling. Also identify any variable(s) that may introduce data leakage or unfairness, and briefly justify your answer.